



**Barker**  
College

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Student Number

**2022**

YEAR 12  
ASSESSMENT BLOCK

# Mathematics Advanced

Staff Involved:

- ALY • PCB • RJW • RAS • LAK\*
- ESP • AXD • GPF • JZT

FRIDAY 01 APRIL 2022

160 copies

## TOPICS COVERED:

|                                            |                    |
|--------------------------------------------|--------------------|
| Year 11 Content                            | Sequences & Series |
| Series & Finance                           | Graphs & Equations |
| Curve Sketching using the Derivative       | Integration        |
| Differentiation of Trigonometric functions |                    |

### General

#### Instructions:

- Working time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working
- Write your student number at the top of every page
- Diagrams not to scale

### Total marks:

80

#### Section I - 8 marks (pages 2 - 4)

- Attempt Multiple Choice Questions 1 - 8
- Allow about 15 minutes for this section

#### Section II - 72 marks (pages 5 - 21)

- Attempt Questions 9 - 16
- Show all necessary working
- Allow about 1 hours and 45 minutes for this section

## Section I

8 marks

Attempt Questions 1 – 8

Allow about 15 minutes for this section

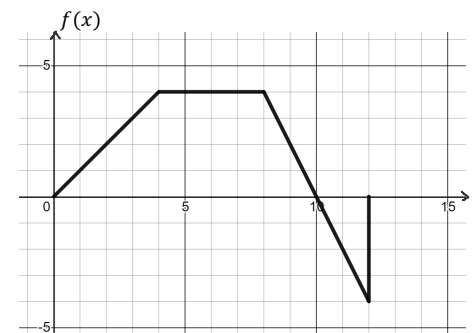
Use the multiple-choice answer sheet for Questions 1 – 8.

- 1 Given that  $\sin \theta = \frac{5}{13}$  and  $\tan \theta < 0$ , what is the value of  $\sec \theta$ ?

- A.  $\frac{13}{5}$
- B.  $-\frac{13}{12}$
- C.  $-\frac{13}{5}$
- D.  $\frac{13}{12}$

- 2 Consider the function,  $f(x)$ , depicted in the graph below.  
Evaluate

$$\int_0^{12} f(x) dx$$

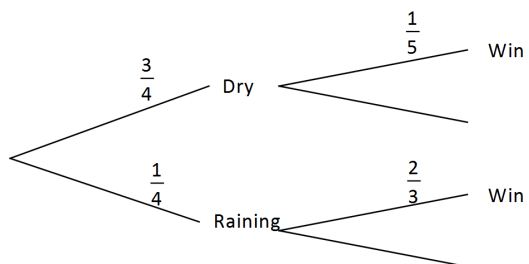


- A. 24
- B. 28
- C. 32
- D. 48

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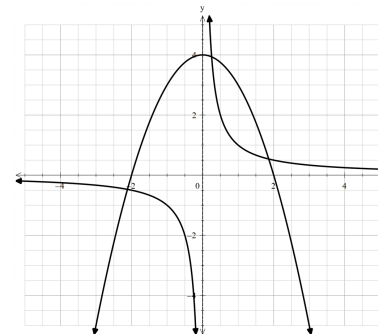
- 3 Which of the following represents the range for the function  $f(x) = e^{2x} - 1$
- A.  $[-1, \infty)$   
 B.  $(-1, \infty)$   
 C.  $(-\infty, -1]$   
 D.  $(-\infty, -1)$
- 4 The manager of a team notices that the team has a probability of  $\frac{2}{3}$  of winning the game if it is raining but, if it is dry, the probability of the team winning is  $\frac{1}{5}$ . The probability that it will rain on a day when they play is  $\frac{1}{4}$ . This information is partially summarised in the tree diagram below.



What is the probability that the team will not win their next game?

- A.  $\frac{13}{15}$   
 B.  $\frac{17}{15}$   
 C.  $\frac{19}{60}$   
 D.  $\frac{41}{60}$
- 5 What is the solution to the following inequation?  
 $x^2 \geq 16$
- A.  $x \geq 4$   
 B.  $-4 \leq x \leq 4$   
 C.  $x \leq -4$  or  $x \geq 4$   
 D.  $-4 \geq x \geq 4$

- 6 Consider the graphs of  $y = \frac{1}{x}$  and  $y = 4 - x^2$  shown below. How many solutions are there to the equation  $\frac{1}{x} = 4 - x^2$ ?



- A. 1 solution  
 B. 2 solutions  
 C. 3 solutions  
 D. 4 solutions
- 7 Given that the limiting sum of the series  $5 - 5x + 5x^2 - \dots$  is 15, what is  $x$ ?
- A.  $x = \frac{2}{3}$   
 B.  $x = -\frac{2}{3}$   
 C.  $x = \frac{2}{15}$   
 D.  $x = -\frac{2}{15}$
- 8 Solve  $2 \sin^2 x - 1 = \sin x$  for  $0 \leq x \leq 2\pi$
- A.  $x = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$   
 B.  $x = \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$   
 C.  $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$   
 D.  $x = \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

End of Section I

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## Section II

72 marks

Attempt Questions 9 – 16

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate space provided.

In Questions 9 – 16, your responses should include relevant mathematical reasoning and/or calculations.

### Question 9 (9 marks)

(a) Solve

2

$$\frac{x+1}{5} - \frac{x-1}{3} = 1$$

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(b) Differentiate

$$\alpha. f(x) = (2x^3 - 4)^5$$

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$$\beta. y = 3e^{2x+5}$$

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### Question 9 (cont.)

(c) Find  $\int 9x^2 - 4 + \frac{3}{x^3} dx$ , simplifying your answer using positive indices.

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(d) Given that  $\log_a 3 = 0.477$  and  $\log_a 5 = 0.699$  evaluate  $\log_a 45$

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(e) A circle is described by the expression  $y^2 - 6y + x^2 + 2x + 2 = 0$ .  
What is the centre and radius of this circle?

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**Question 10** (9 marks)

**(a)** By considering the translations applied, and your knowledge of  $y = \frac{1}{x}$ , sketch

$y = 3 - \frac{1}{x-2}$ , showing important features and the  $x, y$  intercepts.

**3**

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**Question 10** (cont.)

**(b)** Consider a function  $f(x)$ .

(i) Explain the sequence of translations and/or dilations that have been applied to  $f(x)$  to produce  $y = -2f(3x - 12)$ .

**3**

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(ii) The point,  $(-6, 8)$  lies on  $f(x)$ . What are the new co-ordinates of this point after the changes above have been applied?

**2**

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**(c)** In the game of Yahtzee 5 six-sided dice are rolled once.

What is the probability that at least one of the dice is a 6?

**1**

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Student Number

**Question 11** (9 marks)

(a) The discrete probability distribution for an experiment is shown in the table below.

| $x$        | 0     | 1    | 2   | 3    | 4      |
|------------|-------|------|-----|------|--------|
| $P(X = x)$ | $p =$ | 0.15 | 0.3 | 0.25 | $2p =$ |
|            |       |      |     |      |        |
|            |       |      |     |      |        |

- (i) Show that  $p = 0.1$  and hence complete the discrete probability distribution table, assuming no other values of  $x$  are possible. **1**

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- (ii) Find  $E(X)$ . **2**

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- (iii) Calculate the standard deviation, to 2 d.p. **2**

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- (iv) What is the probability that an outcome for this experiment is odd? **1**

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**Question 11** (cont.)

- (v) Given that the outcome for an experiment was odd, what is the probability that the outcome was a 3? **1**

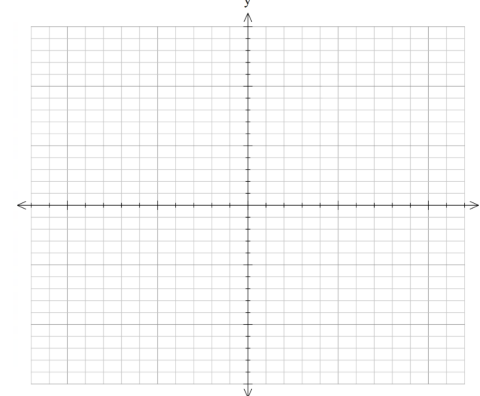
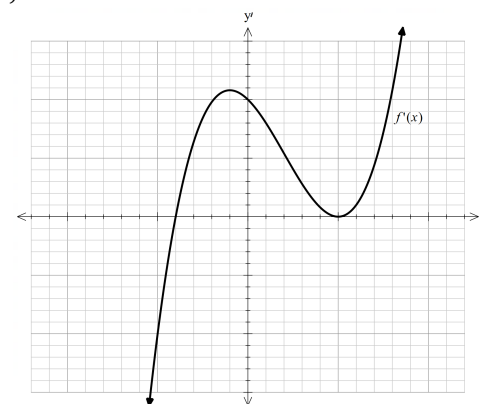
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- (b) The graph below shows the gradient function,  $f'(x)$ . On the grid below sketch a possible curve for  $f(x)$ . **2**



Student Number

(a) Consider the function

$$f(x) = -\frac{x^3}{3} + x^2 + 3x - 9$$

- (i) Find the coordinates of the turning points and determine their nature 3

[illegible]

- (ii) Factorise this function and identify the  $x$  – intercepts. 2

[illegible]

- 11 -

(iii) Hence, sketch the curve, showing all this information, and the  $y$ -intercept. 2

[illegible]

- (b)** The sum of the first 8 terms of an arithmetic series is equal to 18. If the first term is  $-4$ , what is the value of the eighth term? **2**

[illegible]

- 12 -

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Student Number

**Question 13** (9 marks)

- (a) The zoom function in a software package multiplies the dimensions of an image by 1.2.

In an image, the height of a building is 50 mm.

After the zoom function is applied once, the height of the building in the image is 60 mm.

After a second application, its height is 72 mm.

The height of the building in the image is required to be more than 400 mm.

Starting from the original image, what is the least number of times the zoom function must be applied?

3

- (b) Evaluate

3

$$\int_{-3}^0 \sqrt{3x+9} \, dx$$

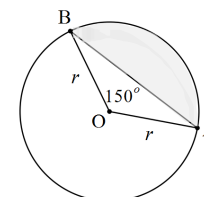
**Question 13** (cont.)

- (c) Consider the diagram below, where a triangle,  $\triangle AOB$ , is inscribed in a circle of radius  $r$ .

The angle between the radii  $OA$  and  $OB$  is  $\angle AOB = 150^\circ$ .

Given that the area of the shaded minor segment is  $16 \, m^2$ , what is the length of the radius (2 d.p.)?

3



Student Number

(a) Consider the curve

$$y = \frac{x^2}{4} + 2x - 4$$

- (i) Find the equation of the normal to the curve at the point on the curve where

$$x = -6$$

3

[illegible]

- (ii) What are the coordinates of the point where the normal crosses the curve a second time? **2**

2

[illegible]

**(b)** Find  $a$  and  $b$  if the curve  $y = x^4 + ax^3 + bx^2$  has an inflection point at  $(2, 0)$ .

4

[illegible]

Student Number

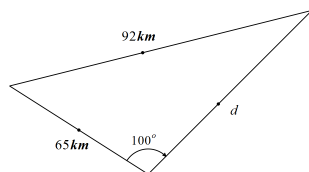
**(a)** Differentiate  $f(x)$ , writing your answer in simplest form.

2

$$f(x) = \frac{x}{\sqrt{x^2 + 1}}$$

[illegible]

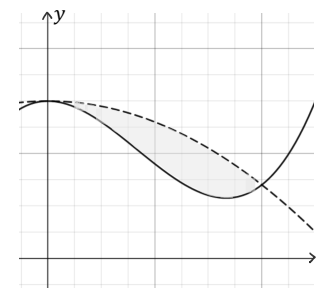
3

[illegible]

- 17 -

(c) The functions,  $f(x) = x^3 - 5x^2 + 30$  and  $g(x) = 30 - x^2$  intersect as shown in the diagram below. Find the area of the shaded region.

4

[illegible]

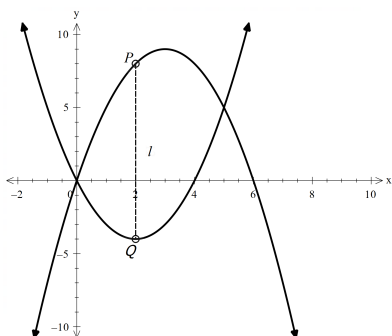
- 18 -

Student Number

(a) In the diagram below, the points  $P$  and  $Q$  are on the curves,  $y = x(6 - x)$  and  $y = x(x - 4)$  respectively.

By showing that the length,  $l$  units, of  $PQ$  is given by  $l = 10x - 2x^2$ , find the maximum length of  $PQ$ , for  $0 \leq x \leq 5$ .

**3**

[illegible]

Future value interest factors for an annuity of \$1 for various interest rates (r) and numbers of periods (n) are given in the table.

|                       |    | Interest Rate per period (r) |         |         |         |         |         |         |         |         |
|-----------------------|----|------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|
|                       |    | 0.5%                         | 1%      | 2%      | 3%      | 4%      | 5%      | 6%      | 7%      | 8%      |
| number of periods (n) | 6  | 6.1059                       | 6.2135  | 6.4343  | 6.6625  | 6.8983  | 7.1420  | 7.3938  | 7.6540  | 7.9228  |
|                       | 7  | 7.1414                       | 7.2857  | 7.5830  | 7.8923  | 8.2142  | 8.5491  | 8.8975  | 9.2598  | 9.6366  |
|                       | 8  | 8.1821                       | 8.3685  | 8.7546  | 9.1591  | 9.5828  | 10.0266 | 10.4913 | 10.9780 | 11.4876 |
|                       | 9  | 9.2280                       | 9.4622  | 9.9497  | 10.4639 | 11.0061 | 11.5779 | 12.1808 | 12.8164 | 13.4866 |
|                       | 10 | 10.2792                      | 10.5668 | 11.1687 | 11.8078 | 12.4864 | 13.2068 | 13.9716 | 14.7836 | 15.6455 |
|                       | 11 | 11.3356                      | 11.6825 | 12.4121 | 13.1920 | 14.0258 | 14.9171 | 15.8699 | 16.8885 | 17.9771 |
|                       | 12 | 12.3972                      | 12.8093 | 13.6803 | 14.6178 | 15.6268 | 16.7130 | 17.8821 | 19.1406 | 20.4953 |

(i) Using the table, calculate the value of Jo's annuity after 10 years.

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$$A_n = 419\,008.5 \times 1.0025^n - 800\,000(1.0025^n - 1)$$

3

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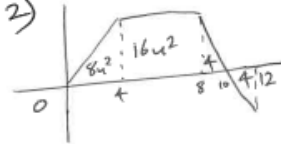
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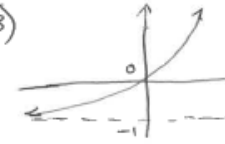
EXTRA WORKING

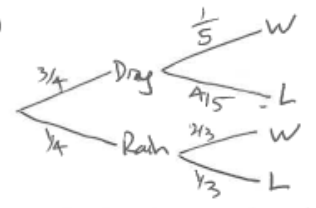
2022 Year 12 Maths Advanced Assessment 2 solutions (1 April)

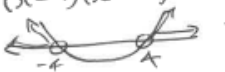
Multiple Choice - Section I

1)   $\sec \theta = \frac{1}{\cos \theta} = \frac{13}{12}$  (B)

2)   $\int_0^{12} f(x) dx = 8 + 16 + 4 = 24$  (A)

3)  Range:  $y > -1$   $(-1, \infty)$  (B)

4)   $P(\text{lose}) = \frac{3}{4} \times \frac{4}{5} + \frac{1}{4} \times \frac{1}{3} = \frac{41}{60}$  (D)

5)  $x^2 - 16 \geq 0$   
 $(x-4)(x+4) \geq 0$   
  $x \leq -4$  or  $x \geq 4$  (C)

6) 3 points of intersection  $\Rightarrow$  3 solutions (C)

7)  $a = 5$   $r = -x$   $\therefore x = \frac{-10}{15} = -\frac{2}{3}$   
 $15 = \frac{5}{1+x}$   
 $15 + 15x = 5$   
 $15x = -10$  (B)

8)  $2 \sin^2 x - \sin x - 1 = 0$   
 $(2 \sin x + 1)(\sin x - 1) = 0$   
 $\sin x = -\frac{1}{2}$  or  $\sin x = 1$   
 $x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{2}$  (C)

Section II

Q9) (a)  $3(x+1) - 5(x-1) = 15$   
 $3x + 3 - 5x + 5 = 15$   
 $-2x + 8 = 15$   
 $-2x = 7$   $\therefore x = -\frac{7}{2}$

(b) (i)  $f'(x) = 5 \cdot (2x^3 - 4)^4 \cdot 6x^2$   
 $= 30x^2 (2x^3 - 4)^4$

(ii)  $\frac{dy}{dx} = 6e^{2x+5}$

(c)  $\int 9x^2 - 4 + 3x^{-3} dx$   
 $= \frac{9x^3}{3} - 4x + \frac{3x^{-2}}{-2} + C$   
 $= 3x^3 - 4x - \frac{3}{2x^2} + C$

(d)  $\log_a 45 = \log_a (9 \times 5)$   
 $= \log_a 9 + \log_a 5$   
 $= 2 \log_a 3 + \log_a 5$   
 $= 2 \times 0.477 + 0.699$   
 $= 1.653$

(e)  $x^2 + 2x + 1 + y^2 - 6y + 9 = -2 + 9 + 1$   
 $(x+1)^2 + (y-3)^2 = 8$   
 centre  $(-1, 3)$   
 Radius  $= \sqrt{8}$  units  
 $= 2\sqrt{2}$  units

$$Q10) (a) f(0) = 3 - \frac{1}{2} = 3\frac{1}{2}$$

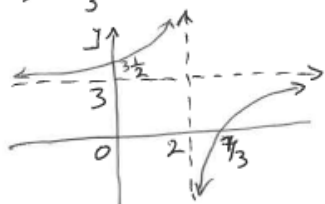
$$0 = 3 - \frac{1}{x-2}$$

$$\frac{1}{x-2} = 3 \quad x \neq 2$$

$$\therefore 1 = 3x - 6 \quad y \neq 3$$

$$3x = 7$$

$$x = \frac{7}{3}$$



$$(b)(i) y = -2f(3(x-4))$$

1. Horizontal stretch by a factor of  $\frac{1}{3}$  i.e.  $f(3x)$

2. Horizontal shift right by 4 units, i.e.  $f(3(x-4))$

3. Vertical stretch by a factor of 2, i.e.  $2f(3(x-4))$

4. Reflection in x-axis i.e.  $-2f(3(x-4))$

$$(ii) 1. (-6, 8) \rightarrow (-2, 8)$$

$$2. (-2, 8) \rightarrow (2, 8)$$

$$3. (2, 8) \rightarrow (2, 16)$$

$$4. (2, 16) \rightarrow (2, -16)$$

$$\therefore (2, -16)$$

$$(c) P(\text{at least one 6}) = 1 - P(\text{no 6s})$$

$$= 1 - \left(\frac{5}{6}\right)^5$$

$$= \frac{4651}{7776}$$

$$Q11) (a) (i) p + 2p + 0.15 + 0.3 + 0.25 = 1$$

$$3p + 0.7 = 1$$

$$3p = 0.3 \quad \therefore p = 0.1$$

| x                    | 0   | 1    | 2   | 3    | 4   |
|----------------------|-----|------|-----|------|-----|
| P(X=x)               | 0.1 | 0.15 | 0.3 | 0.25 | 0.2 |
| x.p(x)               | 0   | 0.15 | 0.6 | 0.75 | 0.8 |
| x <sup>2</sup> .p(x) | 0   | 0.15 | 1.2 | 2.25 | 3.2 |

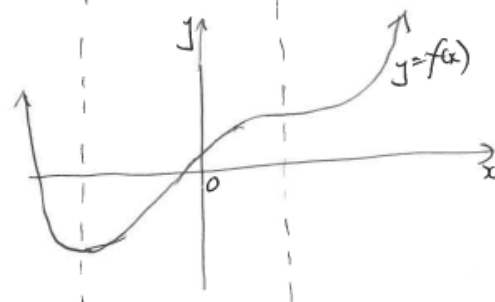
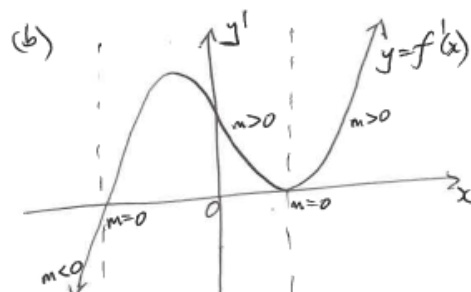
$$(ii) E(X) = 0 + 0.15 + 0.6 + 0.75 + 0.8 = 2.3$$

$$(iii) Var(X) = (0 + 0.15 + 1.2 + 2.25 + 3.2) - 2.3^2 = 1.51$$

$$\sigma = \sqrt{1.51} \approx 1.23$$

$$(iv) P(X=1 \text{ or } X=3) = 0.15 + 0.25 = 0.4$$

$$(v) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.25}{0.4} = \frac{5}{8}$$



$$Q12) (a) f(x) = -\frac{x^3}{3} + x^2 + 3x - 9$$

$$(i) f'(x) = -x^2 + 2x + 3$$

$$f'(x) = 0 \text{ for stat pt}$$

$$\therefore 0 = -x^2 + 2x + 3$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3 \text{ or } -1$$

$$y = 0 \text{ or } -\frac{32}{3}$$

$$f''(x) = -2x + 2$$

$$f''(3) = -6 + 2 < 0 \quad \wedge$$

$\therefore$  Max turning pt @ (3, 0)

$$f''(-1) = 2 + 2 > 0 \quad \vee$$

$\therefore$  Min turning pt @ (-1, -32/3)

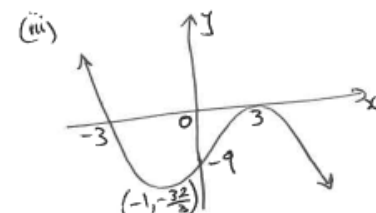
$$(ii) -\frac{1}{3}(x^3 - 3x^2 - 9x + 27) = 0$$

$$x^2(x-3) - 9(x-3) = 0$$

$$(x-3)(x^2 - 9) = 0$$

$$(x-3)(x-3)(x+3) = 0$$

$$\therefore x = 3, 3 \text{ or } -3$$



$$(b) S_8 = 18 \quad a = -4$$

using  $S_n = \frac{n}{2}(a + l)$  gives

$$18 = \frac{8}{2}(-4 + l)$$

$$18 = 4(l - 4)$$

$$18 = 4l - 16$$

$$4l = 34$$

$$l = T_8 = \frac{34}{4} = \frac{17}{2}$$

$$Q13) 50, 60, 72, \dots \text{ G.P. with } a=50, r=1.2$$

$$400 = 50 \times 1.2^{n-1}$$

$$8 = 1.2^{n-1}$$

$$\log 8 = (n-1) \log 1.2$$

$$n-1 = \frac{\log 8}{\log 1.2}$$

$$n = 1 + \frac{\log 8}{\log 1.2} \div 12.4 \text{ i.e. } T_{13} > 400 \text{ mm}$$

$\therefore$  Need 12 zoom applications

$$(b) \int_0^1 (3x+9)^{1/2} dx$$

$$= \left[ \frac{(3x+9)^{3/2}}{3 \times 3/2} \right]_0^1$$

$$= \frac{2}{9} [9^{3/2} - 0^{3/2}]$$

$$= \frac{2}{9} \times 27 = 6$$

$$(c) 150^\circ = 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$$

$$16 = \frac{1}{2} \times r^2 \times \frac{5\pi}{6} - \frac{1}{2} \times r^2 \sin \frac{5\pi}{6}$$

$$32 = \frac{5\pi}{6} r^2 - \frac{r^2}{2}$$

$$32 \times 6 = 5\pi r^2 - 3r^2$$

$$r^2(5\pi - 3) = 192$$

$$r^2 = \frac{192}{5\pi - 3}$$

$$r = \sqrt{\frac{192}{5\pi - 3}}$$

$$\approx 3.89 \text{ m}$$

$$Q14) (a) y = \frac{x^2}{4} + 2x - 4$$

$$(i) \frac{dy}{dx} = \frac{x}{2} + 2$$

$$f'(-6) = \frac{-6}{2} + 2 = -3 + 2 = -1$$

$$\therefore \text{grad of normal} = 1$$

$$f(-6) = \frac{36}{4} - 12 - 4 = 9 - 12 - 4 = -7 \therefore (-6, -7)$$

$$\text{Eqn of normal is}$$

$$y + 7 = 1(x + 6)$$

$$\therefore y = x - 1$$

$$(ii) y = x - 1 \quad \left\{ \begin{array}{l} x - 1 = \frac{x^2}{4} + 2x - 4 \\ y = \frac{x^2}{4} + 2x - 4 \end{array} \right. \therefore 4x - 4 = x^2 + 8x - 16$$

$$\therefore x^2 + 4x - 12 = 0$$

$$\therefore (x+6)(x-2) = 0$$

$$x = -6 \text{ or } 2$$

$$y = -7 \text{ or } 1 \therefore (2, 1)$$

$$(b) f(2) = 0 \therefore 0 = 16 + 8a + 4b$$

$$\therefore 8a + 4b = -16$$

$$2a + b = -4 \quad \text{--- (1)}$$

$$\frac{dy}{dx} = 4x^3 + 3ax^2 + 2bx$$

$$\frac{d^2y}{dx^2} = 12x^2 + 6ax + 2b$$

$$f''(2) = 0 \therefore 0 = 48 + 12a + 2b$$

$$12a + 2b = -48$$

$$6a + b = -24 \quad \text{--- (2)}$$

$$2a + b = -4 \quad \left\{ \begin{array}{l} 4a = -20 \\ 6a + b = -24 \end{array} \right. \therefore a = -5$$

$$6a + b = -24$$

$$\therefore -10 + b = -4$$

$$b = 6$$

$$\therefore a = -5, b = 6$$

$$Q15) f(x) = \frac{x}{\sqrt{x^2+1}} = \frac{u}{v}$$

$$(a) f'(x) = \frac{\sqrt{x^2+1} \cdot 1 - x \cdot \frac{1}{2} \cdot (x^2+1)^{-1/2} \cdot 2x}{(\sqrt{x^2+1})^2}$$

$$= \frac{(\sqrt{x^2+1} - \frac{x^2}{\sqrt{x^2+1}})}{x^2+1} \times \frac{\sqrt{x^2+1}}{\sqrt{x^2+1}} = \frac{x^2+1 - x^2}{(x^2+1)\sqrt{x^2+1}} = \frac{1}{(x^2+1)\sqrt{x^2+1}}$$

$$(b) q_2^2 = 65^2 + d^2 - 2 \times 65d \cos 100^\circ$$

$$\therefore d^2 - 130 \cos 100^\circ d - 4239 = 0$$

$$d = \frac{+130 \cos 100^\circ \pm \sqrt{(130 \cos 100^\circ)^2 - 4 \times 1 \times -4239}}{2}$$

$$= \frac{130 \cos 100^\circ \pm \sqrt{(130 \cos 100^\circ)^2 + 16956}}{2}$$

$$\therefore d = \frac{130 \cos 100^\circ + \sqrt{(130 \cos 100^\circ)^2 + 16956}}{2} \div 55 \text{ km}$$

$$(c) x^3 - 5x^2 + 30 = 30 - x^2$$

$$x^3 - 4x^2 = 0$$

$$x^2(x-4) = 0$$

$$\therefore x = 0 \text{ or } 4$$

$$A = \int_0^4 (30 - x^2) - (x^3 - 5x^2 + 30) dx$$

$$= \int_0^4 30 - x^2 - x^3 + 5x^2 - 30 dx$$

$$= \int_0^4 4x^2 - x^3 dx$$

$$= \left[ \frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4$$

$$= \left[ \frac{4 \times 4^3}{3} - \frac{4^4}{4} - (0 - 0) \right]$$

$$= \frac{64}{3} \text{ units}^2$$

$$Q16) (a) l = 6x - x^2 - (x^2 - 4x) = 6x - x^2 - x^2 + 4x$$

$$\therefore l = 10x - 2x^2$$

$$\frac{dl}{dx} = 10 - 4x$$

$$10 - 4x = 0$$

$$4x = 10 \therefore x = \frac{10}{4} = \frac{5}{2}$$

$$\frac{d^2l}{dx^2} = -4 < 0 \therefore \text{Max when } x = \frac{5}{2}$$

$$\therefore \text{Max length of PQ} = 10 \times \frac{5}{2} - 2 \times \left(\frac{5}{2}\right)^2 = 12.5 \text{ units}$$

$$(b) 6\% \text{ for 10 periods} = 13.9716$$

$$\therefore 13.9716 \times 5000 = \$69858$$

$$(ii) A_1 = 419008.50 \times 1.0025 - 2000$$

$$A_2 = A_1 \times 1.0025 - 2000$$

$$= 419008.50 \times 1.0025^2 - 2000 \times 1.0025 - 2000$$

$$A_3 = A_2 \times 1.0025 - 2000$$

$$= 419008.50 \times 1.0025^3 - 2000 \times 1.0025^2 - 2000 \times 1.0025 - 2000$$

$$A_n = 419008.50 \times 1.0025^n - 2000 \times 1.0025^{n-1} - \dots - 2000 \times 1.0025 - 2000$$

$$= 419008.50 \times 1.0025^n - 2000(1 + 1.0025 + \dots + 1.0025^{n-1})$$

$$\text{A.P. with } a=1, r=1.0025, n=n$$

$$= 419008.50 \times 1.0025^n - 2000 \times \frac{(1.0025^n - 1)}{1.0025 - 1}$$

$$= 419008.50 \times 1.0025^n - \frac{2000}{0.0025} (1.0025^n - 1)$$

$$A_n = 419008.50 \times 1.0025^n - 800000 (1.0025^n - 1)$$

$$(iii) A_n = 0 \therefore 0 = 419008.50 \times 1.0025^n - 800000 \times 1.0025^n + 800000$$

$$0 = -380991.50 \times 1.0025^n + 800000$$

$$380991.50 \times 1.0025^n = 800000$$

$$1.0025^n = \frac{800000}{380991.50}$$

$$1.0025^n = 2.099784 \dots$$

$$n \log 1.0025 = \log 2.099784 \dots$$

$$n = \frac{\log 2.099784 \dots}{\log 1.0025}$$

$$n \div 297.1 \text{ months}$$